1 Fundamental Equations

1.1 Mass Balance ODE

$$\frac{d\mathbf{S}}{dt} = \mathbf{N} \cdot \mathbf{v}(\mathbf{S}, \mathbf{p})$$

- $\mathbf{S}$ = concentration vector (species)
- $\mathbf{N}$ = stoichiometry matrix
- $\mathbf{v}$ = rate vector (fluxes)
- $\mathbf{p}$ = parameter vector

1.2 Stoichiometry Matrix Properties

- Rows = species (m), Columns = reactions (r)
- Rank($\mathbf{N}$) = m_0 = # independent species
- $m - m_0$ = # conservation relations

1.3 Steady State

At steady state: $\mathbf{N} \cdot \mathbf{v}^0 = \mathbf{0}$

2 Stoichiometric Analysis

2.1 Null Spaces of N

Right null space (kernel):

$$\mathbf{N} \cdot \mathbf{K} = \mathbf{0}$$

- Columns of $\mathbf{K}$ = steady-state flux vectors
- Dimension = $r - m_0$ = degrees of freedom
- Any steady-state flux: $\mathbf{v}^{ss} = \mathbf{K} \cdot \boldsymbol{\alpha}$

Left null space:

$$\mathbf{G} \cdot \mathbf{N} = \mathbf{0}$$

- Rows of $\mathbf{G}$ = conservation relations
- Dimension = $m - m_0$
- Conserved quantity: $\mathbf{G} \cdot \mathbf{S} = \mathbf{T}$ (constant)

2.2 The Link Matrix L

Partition species into **independent** ($\mathbf{S}_i$) and **dependent** ($\mathbf{S}_d$):

$$\mathbf{S} = \begin{pmatrix} \mathbf{S}_i \\ \mathbf{S}_d \end{pmatrix}$$

The **link matrix L** relates them:

$\mathbf{S} = \mathbf{L}\mathbf{S}_i + \mathbf{T}$, where $\mathbf{L} = \begin{pmatrix} \mathbf{I} \\ \mathbf{L}_0 \end{pmatrix}$

- $\mathbf{L}_0 = (m - m_0) \times m_0$ matrix
- $\mathbf{T}$ = conserved totals vector
- $\mathbf{L}_0$ comes from row reduction of $\mathbf{N}$

2.3 Reduced Stoichiometry Matrix

$$\boxed{\mathbf{N} = \mathbf{L} \cdot \mathbf{N}_r}$$

- $\mathbf{N}_r$ is $m_0 \times r$ (full row rank)
- Rows correspond to independent species only
- Reduced system: $\frac{d\mathbf{S}_i}{dt} = \mathbf{N}_r \cdot \mathbf{v}$

2.4 Relationships Between Matrices

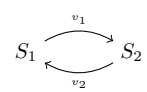
$$\mathbf{N} = \mathbf{L} \cdot \mathbf{N}_r$$
$$\mathbf{G} \cdot \mathbf{L} = \mathbf{0}$$
$$\mathbf{L}^T \cdot \mathbf{G}^T = \mathbf{0}$$

2.5 Computing L from N

Method: Row reduce $\mathbf{N}$ to row echelon form

1. Identify pivot columns → independent species
2. Non-pivot rows give $\mathbf{L}_0$
3. $\mathbf{L} = \begin{pmatrix} \mathbf{I} \\ \mathbf{L}_0 \end{pmatrix}$

2.6 Example: Conserved Cycle



Stoichiometry matrix:

$$\mathbf{N} = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{rank} = 1$$

- One independent species (e.g., S_1)
- Conservation: $S_1 + S_2 = T$ (constant)

Link matrix:

$$\mathbf{L} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{L}_0 = (-1)$$

So $S_2 = -S_1 + T = T - S_1$

Reduced system: $\mathbf{N}_r = (-1 \quad 1)$

$\frac{dS_1}{dt} = -v_1 + v_2$

2.7 The K Matrix (Kernel)

$\mathbf{K}$ spans all possible steady-state fluxes:

$$\mathbf{v}^{ss} = \mathbf{K} \cdot \mathbf{j}$$

where $\mathbf{j}$ = vector of independent fluxes

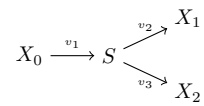
Computing K:

1. Row reduce $\mathbf{N}^T$
2. Identify free variables
3. Express basic variables in terms of free
4. Columns of $\mathbf{K}$ = basis vectors

Dimension check:

$$\dim(\mathbf{K}) = r - \text{rank}(\mathbf{N}) = r - m_0$$

2.8 Example: Branched Pathway



X_0, X_1, X_2 = external (fixed); S = internal variable

Stoichiometry matrix:

$$\mathbf{N} = \begin{pmatrix} 1 & -1 & -1 \end{pmatrix} \quad (1 \times 3)$$

- rank($\mathbf{N}$) = 1, so $\dim(\mathbf{K}) = 3 - 1 = 2$

Kernel matrix:

$$\mathbf{K} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Columns represent independent flux patterns:

- $(1, 1, 0)^T$: flux through branch 1
- $(1, 0, 1)^T$: flux through branch 2

Any steady-state flux: $\mathbf{v} = \alpha \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

At steady state: $v_1 = v_2 + v_3$ (branch point balance)

3 Elementary Modes

3.1 Definition

An **elementary mode (EM)** is a minimal set of reactions that can operate at steady state.

Properties:

- Satisfies $\mathbf{N} \cdot \mathbf{e} = \mathbf{0}$
- Non-decomposable (no subset is valid)
- Unique up to scaling

3.2 Relation to K Matrix

- Columns of $\mathbf{K}$ span the flux space
- EMs are the *edges* of the flux cone
- Every steady-state flux is a non-negative combination of EMs (for irreversible reactions)

3.3 Elementary Mode Matrix E

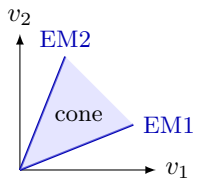
$$\mathbf{N} \cdot \mathbf{E} = \mathbf{0}$$

Each column of $\mathbf{E}$ is an elementary mode.

Any feasible flux:

$$\mathbf{v} = \mathbf{E} \cdot \boldsymbol{\omega}, \quad \boldsymbol{\omega} \geq \mathbf{0}$$

3.4 Flux Cone Geometry



- EMs define edges of the cone
- Interior points = combinations of EMs
- Reversible reactions: cone becomes subspace

3.5 Finding Elementary Modes

Simple cases: Inspect network for minimal cycles

Algorithm (conceptual):

1. Start with null space basis $\mathbf{K}$
2. Find extreme rays of $\{\mathbf{v} : \mathbf{N}\mathbf{v} = 0, \mathbf{v} \geq 0\}$
3. Each extreme ray = one EM

Software: `efmtool`, `metatool`, `COPASI`

3.6 Biological Interpretation

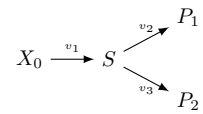
- EMs = "pathways through the network"
- Each EM is a functional unit
- Cell uses weighted sum of EMs
- Minimal: removing any reaction breaks it

3.7 Example: Linear Pathway

$$X_0 \xrightarrow{v_1} S_1 \xrightarrow{v_2} S_2 \xrightarrow{v_3} X_1$$

- $\mathbf{N} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix}$
- Only one EM: $\mathbf{e} = (1, 1, 1)^T$
- All fluxes equal at steady state

3.8 Example: Branched Pathway



- Two EMs: $(1, 1, 0)^T$ and $(1, 0, 1)^T$
- Branch point creates alternatives

4 Elasticities

4.1 Definition

Unscaled elasticity:

$$\varepsilon_{ij}^{\text{unscaled}} = \frac{\partial v_i}{\partial S_j}$$

Scaled elasticity:

$$\boxed{\varepsilon_{ij} = \frac{S_j}{v_i} \cdot \frac{\partial v_i}{\partial S_j} = \frac{\partial \ln v_i}{\partial \ln S_j}}$$

4.2 Common Elasticity Formulas

Michaelis-Menten: $v = \frac{V_{\max} S}{K_m + S}$

$$\varepsilon_S = \frac{K_m}{K_m + S} \quad \text{(ranges 0 to 1)}$$

Reversible MM: $v = \frac{V_f(S/K_S) - V_r(P/K_P)}{1 + S/K_S + P/K_P}$

$$\varepsilon_S = \frac{1 + P/K_P}{1 + S/K_S + P/K_P} \cdot \frac{v_f}{v}$$

$$\varepsilon_P = -\frac{1 + S/K_S}{1 + S/K_S + P/K_P} \cdot \frac{v_r}{v}$$

Hill kinetics: $v = \frac{V_{\max} S^n}{K^n + S^n}$

$$\varepsilon_S = n \cdot \frac{K^n}{K^n + S^n} \quad \text{(ranges 0 to } n\text{)}$$

Mass action: $v = k \cdot S$

$$\varepsilon_S = 1$$

Inhibition: $v = \frac{V_{\max} S}{(K_m + S)(1 + I/K_i)}$

$$\varepsilon_I = -\frac{I/K_i}{1 + I/K_i} \quad \text{(negative, ranges } -1 \text{ to } 0\text{)}$$

5 Linearization

5.1 Taylor Expansion

Around steady state $(\mathbf{S}^0, \mathbf{v}^0)$:

$$\frac{d(\delta \mathbf{S})}{dt} = \mathbf{A} \cdot \delta \mathbf{S} + \mathbf{B} \cdot \delta \mathbf{u}$$

5.2 The Jacobian

$$\boxed{\mathbf{A} = \mathbf{N} \cdot \boldsymbol{\varepsilon}^{\text{unscaled}}}$$

Key insight: Jacobian = Structure × Local kinetics

5.3 Jacobian Elements

$$A_{ij} = \sum_k N_{ik} \cdot \frac{\partial v_k}{\partial S_j}$$

6 Transfer Functions

6.1 Laplace Transform

$$s \cdot \delta \mathbf{S}(s) = \mathbf{A} \cdot \delta \mathbf{S}(s) + \mathbf{B} \cdot \delta \mathbf{u}(s)$$

6.2 Transfer Function Definition

$$\boxed{\mathbf{G}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}}$$

6.3 For 2×2 Systems

If $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then:

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{\Delta(s)} \begin{pmatrix} s - d & b \\ c & s - a \end{pmatrix}$$

where $\Delta(s) = (s - a)(s - d) - bc$

6.4 Characteristic Polynomial

$$\Delta(s) = \det(s\mathbf{I} - \mathbf{A}) = s^2 - \operatorname{tr}(\mathbf{A})s + \det(\mathbf{A})$$

6.5 Poles and Zeros

- **Poles** = roots of denominator = eigenvalues of $\mathbf{A}$
- **Zeros** = roots of numerator
- Poles determine *stability*
- Zeros determine *response shape*

7 Eigenvalues & Stability

7.1 2D System Eigenvalues

For $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

$$\lambda_{1,2} = \frac{(a+d) \pm \sqrt{(a+d)^2 - 4(ad-bc)}}{2}$$

Or: $\lambda_{1,2} = \frac{\mathrm{tr}(\mathbf{J}) \pm \sqrt{\mathrm{tr}(\mathbf{A})^2 - 4\det(\mathbf{A})}}{2}$

7.2 Stability Criteria

System is **stable** iff all eigenvalues have $\mathrm{Re}(\lambda) < 0$

For 2D systems:

- Stable iff $\mathrm{tr}(\mathbf{A}) < 0$ AND $\det(\mathbf{A}) > 0$

7.3 Types of Dynamics

Eigenvalues	Behavior
Real, both < 0	Stable node
Real, both > 0	Unstable node
Real, opposite signs	Saddle (unstable)
Complex, Re < 0	Stable spiral
Complex, Re > 0	Unstable spiral
Complex, Re $= 0$	Center (oscillations)

7.4 Time Constants

$$\tau_i = -\frac{1}{\mathrm{Re}(\lambda_i)}$$

Largest τ = slowest mode = dominant dynamics

8 DC Gain & Control Coefficients

8.1 The Key Connection

DC gain $= \mathbf{G}(0) = -\mathbf{A}^{-1}\mathbf{B}$

This equals the **steady-state sensitivity**

8.2 Control Coefficients

Concentration control coefficient:

$$C^{S_i}_{E_j} = \frac{E_j}{S_i} \cdot \frac{\partial S_i}{\partial E_j} = \frac{\partial \ln S_i}{\partial \ln E_j}$$

Flux control coefficient:

$$C^{J}_{E_j} = \frac{E_j}{J} \cdot \frac{\partial J}{\partial E_j} = \frac{\partial \ln J}{\partial \ln E_j}$$

8.3 Summation Theorems

$$\sum_j C^{J}_{E_j} = 1 \quad \text{(flux)}$$

$$\sum_j C^{S_i}_{E_j} = 0 \quad \text{(concentration)}$$

8.4 Branch Point Theorems

For a branch: $v_1 \to S \to v_2$ and $S \to v_3$

Define flux fractions: $\alpha = v_2/v_1$, $\beta = v_3/v_1$

(Note: $\alpha + \beta = 1$ at steady state)

Flux branch point theorems:

$$C^{J_1}_{e_2}(1-\alpha) - C^{J_1}_{e_3}(\alpha) = 0$$
$$C^{J_2}_{e_1}(1-\alpha) + C^{J_2}_{e_3}(\alpha) = 0$$
$$C^{J_3}_{e_1}(\alpha) + C^{J_3}_{e_2} = 0$$

Concentration branch point theorems:

$$C^{S_1}_{e_2}(1-\alpha) + C^{S_1}_{e_3}(\alpha) = 0$$
$$C^{S_1}_{e_1}(1-\alpha) + C^{S_1}_{e_3} = 0$$
$$C^{S_1}_{e_1}(\alpha) + C^{S_1}_{e_2} = 0$$

Combined with summation and connectivity theorems, these allow solving for all control coefficients in branched systems.

Key result: Control coefficients can exceed 1 at branch points (the "branchpoint effect").

Only enzymes in or downstream of branches i or j can control the ratio J_i/J_j.

8.5 Connectivity Theorems

$$\sum_i C^{J}_{E_i} \cdot \varepsilon^{S_k}_i = 0$$

$$\sum_i C^{S_j}_{E_i} \cdot \varepsilon^{S_k}_i = -\delta_{jk}$$

8.6 Matrix Form

$$\mathbf{C} \cdot \boldsymbol{\varepsilon} = -\mathbf{I}$$

Therefore: $\mathbf{C} = -\boldsymbol{\varepsilon}^{-1}$ (when invertible)

This applies to linear chains.

9 Frequency Response

9.1 Sinusoidal Input

For input $u(t) = A\sin(\omega t)$, steady-state output is:

$$y(t) = A|G(j\omega)|\sin(\omega t + \phi)$$

where $\phi = \angle G(j\omega)$

9.2 Bode Plot

- **Magnitude:** $|G(j\omega)|$ vs ω (log-log)
- **Phase:** $\angle G(j\omega)$ vs ω (semilog)

9.3 Computing Magnitude

For $G(j\omega) = \frac{N(j\omega)}{D(j\omega)}$:

$$|G(j\omega)| = \frac{|N(j\omega)|}{|D(j\omega)|}$$

where $|a + jb| = \sqrt{a^2 + b^2}$

9.4 First-Order System

$$G(s) = \frac{K}{1 + \tau s}$$

- DC gain: $|G(0)| = K$
- Corner frequency: $\omega_c = 1/\tau$
- High frequency: $|G| \approx K/(\tau\omega)$ (slope -1)
- Phase: from $0°$ to $-90°$

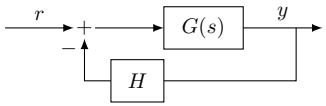
9.5 Second-Order System

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- ω_n = natural frequency
- ζ = damping ratio
- $\zeta < 1$: oscillatory (underdamped)
- $\zeta = 1$: critically damped
- $\zeta > 1$: overdamped

10 Feedback Systems

10.1 Closed-Loop Transfer Function



$$G_{\text{closed}}(s) = \frac{G(s)}{1 + G(s)H(s)}$$

10.2 Loop Gain

$$L(s) = G(s)H(s)$$

- $|L| \gg 1$: output $\approx r/H$ (feedback dominates)
- $|L| \ll 1$: output $\approx G \cdot r$ (open loop)

10.3 Feedback Effects

Property	Effect of Neg. Feedback
Gain	Reduced by $(1 + L)$
Bandwidth	Increased
Disturbance rejection	Improved
Sensitivity to G	Reduced
Stability	Can destabilize

10.4 Effect on Control Coefficients

Negative feedback *redistributes* control:

- Reduces control of feedback-regulated enzyme
- Increases control of the enzyme beyond the signal
- Total still sums to 1 (for flux)

11 Perfect Adaptation

11.1 Definition

Output returns to *same* steady state regardless of sustained input change.

11.2 Transfer Function Requirement

Perfect adaptation $\Leftrightarrow G(0) = 0$

Requires a **zero at** $s = 0$ in the transfer function

11.3 Implementation: Integral Feedback

Minimal system for perfect adaptation:

$$\frac{dY}{dt} = k_a L - k_b M Y$$

$$\frac{dM}{dt} = k_1 Y - k_0$$

- L = input signal (perturbation here)
- Y = output (adapting variable)
- M = adaptation variable (integrator)
- k_0 = constant removal of M (zero-order)

Steady-state analysis:

From $dM/dt = 0$:

$$Y^{ss} = \frac{k_0}{k_1} \quad \text{(independent of } L\text{)}$$

From $dY/dt = 0$:

$$M^{ss} = \frac{k_a L}{k_b Y^{ss}} = \frac{k_a k_1}{k_b k_0} L$$

M adjusts to absorb changes in L, keeping Y at its setpoint.

Key: The constant k_0 (zero-order in M) makes M a true integrator.

11.4 The Integral Controller

The integrator $\frac{1}{s}$ in the feedback path creates:

- A pole at $s = 0$ in $H(s)$
- A zero at $s = 0$ in $G_{\text{closed}}(s)$
- $\Rightarrow$ Perfect adaptation

11.5 High-Pass Filter Behavior

Perfect adaptation means:

- Low frequencies: attenuated ($|G| \to 0$)
- High frequencies: pass through
- System responds to *changes*, ignores *levels*

12 Retroactivity

12.1 Definition

When modules are connected, downstream modules *load* upstream modules, changing their behavior.

12.2 Insulation Strategies

- High-gain cascades (like MAPK)
- Phosphorylation cycles with fast kinetics - debatable
- Enzymatic "push-pull" amplifiers

13 Stability Analysis

13.1 Routh-Hurwitz (2nd Order)

For $\Delta(s) = s^2 + as + b$:

- Stable iff $a > 0$ AND $b > 0$

13.2 Routh-Hurwitz (3rd Order)

For $\Delta(s) = s^3 + as^2 + bs + c$:

- Stable iff $a > 0$, $c > 0$, AND $ab > c$

13.3 Hopf Bifurcation

Oscillations emerge when complex eigenvalues cross imaginary axis:

- $\text{Re}(\lambda) = 0$, $\text{Im}(\lambda) = \omega \neq 0$
- Oscillation period $T = 2\pi/\omega$

13.4 Conditions for Oscillations

1. Negative feedback (delayed)
2. Sufficient loop gain
3. Sufficient delay (or chain length)

14 Useful Identities

14.1 Matrix Inverse (2×2)

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

14.2 Complex Number Magnitude

$$|a + jb| = \sqrt{a^2 + b^2}$$

14.3 Complex Number Phase

$$\angle(a + jb) = \arctan(b/a)$$

14.4 Laplace Transform Pairs

$f(t)$	$F(s)$
e^{-at}	$\frac{1}{s+a}$
$1 - e^{-at}$	$\frac{a}{s(s+a)}$
te^{-at}	$\frac{1}{(s+a)^2}$
$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
step	$\frac{1}{s}$

14.5 Final Value Theorem

If $\lim_{t\to\infty} y(t)$ exists:

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} s \cdot Y(s)$$

15 Quick Reference: Tellurium

15.1 Basic Operations

```
import tellurium as te
r = te.loada('''
  -> S; k1*L - k2*S
  S = 1; k1 = 1; k2 = 0.1; L = 1
''')

# Steady state
r.steadyState()

# Simulate
result = r.simulate(0, 100, 500)
```

15.2 Stoichiometric Analysis

```
# Full stoichiometry matrix N
N = r.getFullStoichiometryMatrix()

# Reduced stoichiometry matrix Nr
Nr = r.getReducedStoichiometryMatrix()

# Link matrix L
L = r.getLinkMatrix()

# Conservation matrix G (left null)
G = r.getConservationMatrix()

# Kernel matrix K (right null)
K = r.getKMatrix()
```

15.3 MCA and Dynamics

```
# Jacobian
A = r.getFullJacobian()

# Eigenvalues
import numpy as np
eigs = np.linalg.eigvals(J)

# Elasticities (scaled)
ee = r.getScaledElasticityMatrix()

# Control coefficients
ccJ = r.getScaledFluxControlCoefficientMatrix()
ccS = r.getScaledConcentrationControlCoefficientMatrix()
```

16 Matrix Summary Table

Matrix	Size	Sym.	Meaning
Stoichiometry	$m \times r$	$\mathbf{N}$	Network structure
Reduced stoich.	$m_0 \times r$	$\mathbf{N}_r$	Indep. species
Link	$m \times m_0$	$\mathbf{L}$	Dep./indep. relation
Kernel	$r \times (r{-}m_0)$	$\mathbf{K}$	SS flux basis
Conservation	$(m{-}m_0) \times m$	$\mathbf{G}$	Conserved moieties
Elem. modes	$r \times n_{EM}$	$\mathbf{E}$	Minimal pathways
Elasticity	$r \times m$	$\boldsymbol{\varepsilon}$	Local kinetics
Jacobian	$m \times m$	$\mathbf{A}$	Dynamics
Control (flux)	$r \times r$	$\mathbf{C}^J$	Flux sensitivity
Control (conc.)	$m \times r$	$\mathbf{C}^S$	Conc. sensitivity

Key: m = species, r = rxns, $m_0 = \text{rank}(\mathbf{N})$

16.1 Fundamental Relationships

$$\mathbf{N} = \mathbf{L} \cdot \mathbf{N}_r \qquad \mathbf{N} \cdot \mathbf{K} = \mathbf{0}$$

$$\mathbf{G} \cdot \mathbf{N} = \mathbf{0} \qquad \mathbf{G} \cdot \mathbf{L} = \mathbf{0}$$

$$\mathbf{J} = \mathbf{N} \cdot \boldsymbol{\varepsilon}^u \qquad \mathbf{C} \cdot \boldsymbol{\varepsilon} = -\mathbf{I}$$

Advanced Control of Biochemical Networks.
Herbert M Sauro March 19, 2026